

CIS 5800

Machine Perception

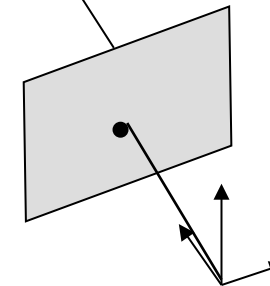
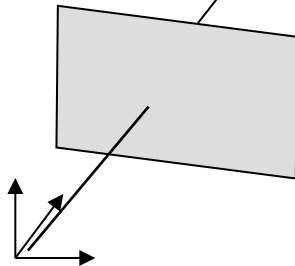
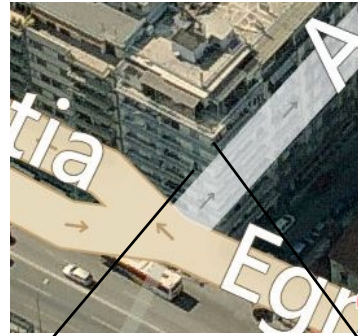
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Lec 10 (Part 2): Feb 24, 2026

3D Motion from Two Views or Structure from Motion (SfM)

Input: Two Calibrated Views of the same 3D scene

Intrinsics known,
So we can always
stick to
“calibrated
coordinates” like
we used in P3P,
rather than pixel
coordinates.



Recap: Pixels → “Calibrated coordinates”

RGB images only provide pixel coordinates u, v for each vertex, not camera-centric 3D coordinates. Convert these to:

“Calibrated coordinates”: $\mathbf{p}_i \sim K^{-1}(u_i \quad v_i \quad 1)^T$

These are essentially image plane coordinates with principal point as origin, and focal length set to 1.

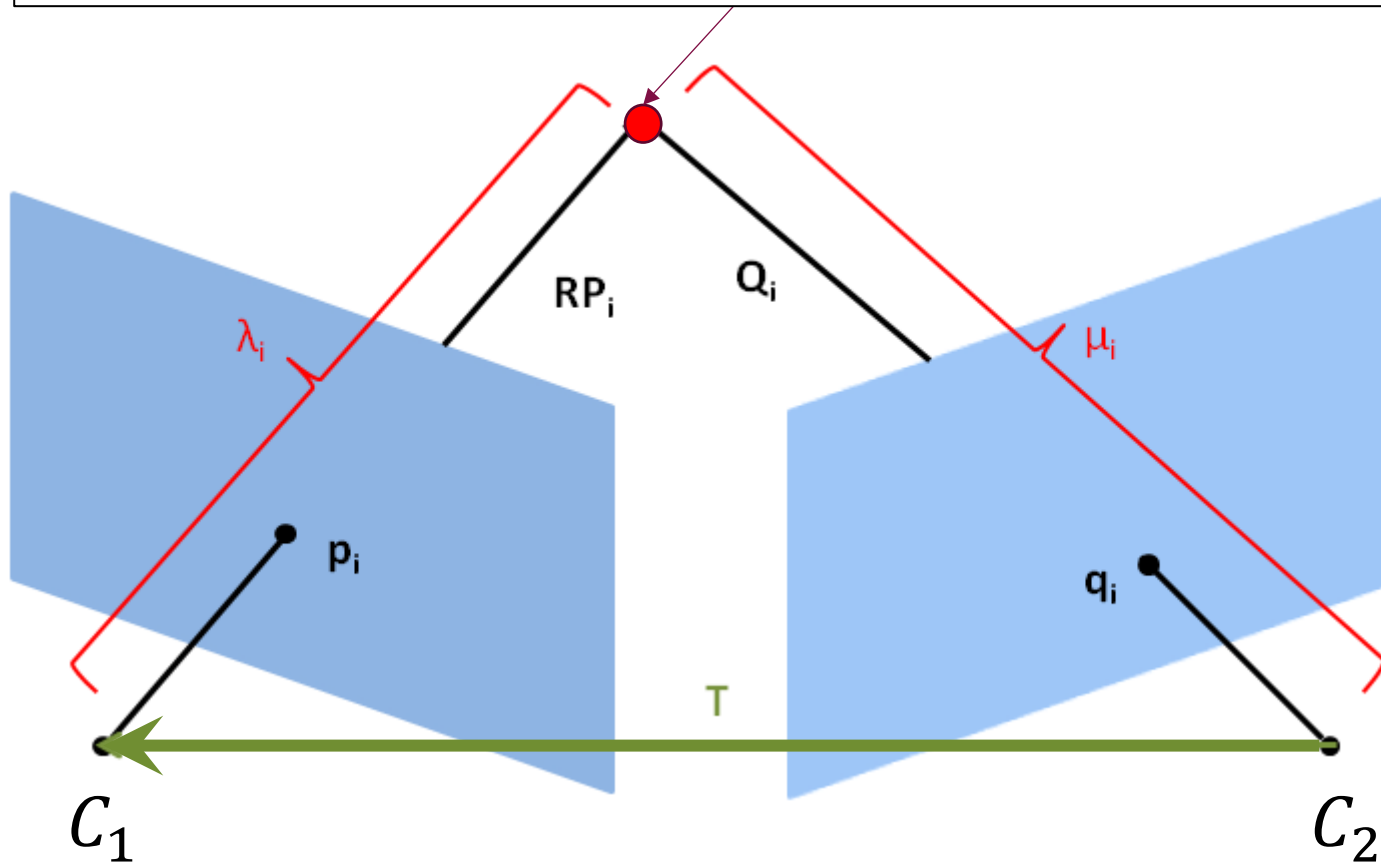
Euclidean interpretation: $\mathbf{p}_i$ is a vector in camera coordinates, originating from the camera center and pointing toward the 3D point corresponding to pixel coordinates (u_i, v_i) .

Recap: Calibrated coordinates + distance = Camera-centric 3D

- After writing as calibrated coordinates, camera-centric 3D coordinates is just one step away: finding the distance of the 3D point along that direction, i.e. “depth”
- To reduce P3P to 3D->3D as we had planned, we will need to find those depths. Indeed, that is the first step of P3P.

Input: Two Calibrated Views of the Same 3D Scene

$[\lambda p]$ in C_1 camera coords or $[\mu q]$ in C_2 camera coords



$$[X]_{C_2} = R[X]_{C_1} + T$$

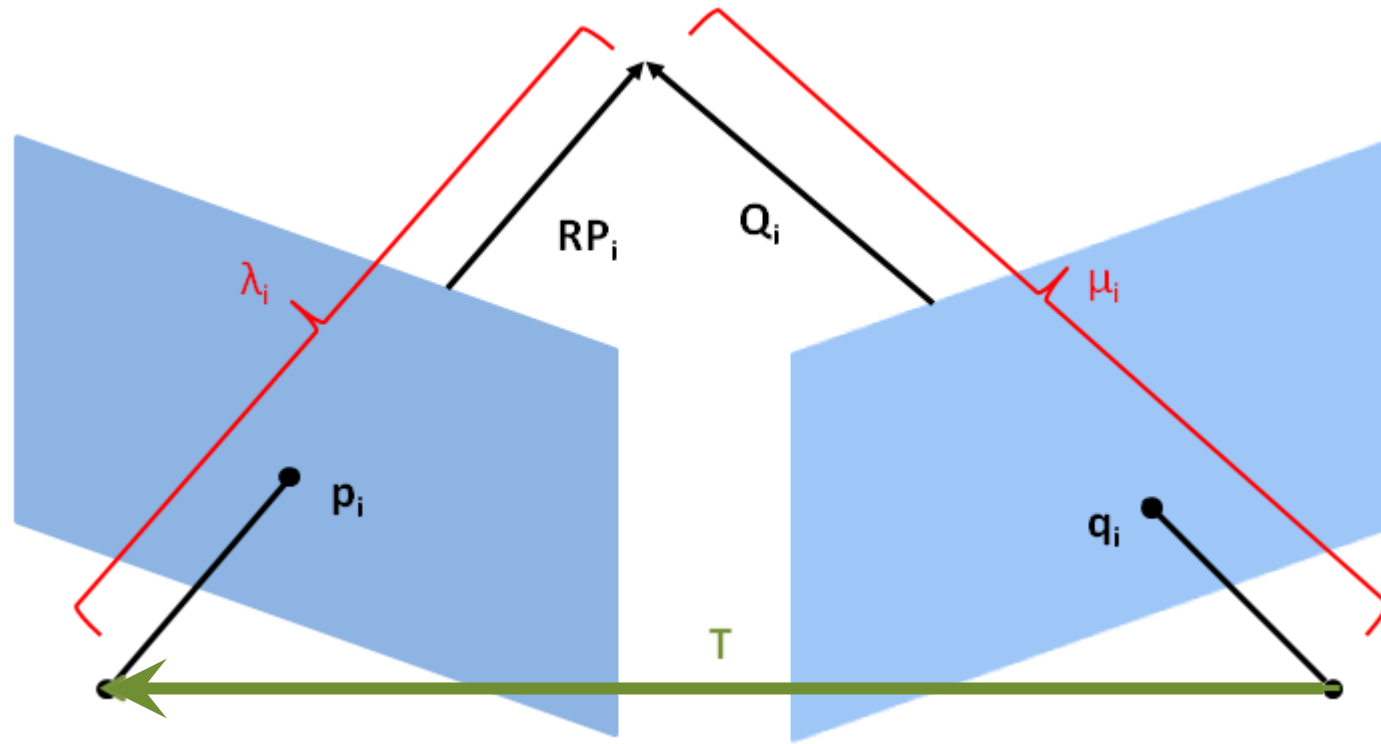
Now, any 3D coordinates in C_1 frame:
 $[X]_{C_1} \rightarrow [R[X]_{C_1} + T]_{C_2}$

So $[\lambda p]_{C_1} \rightarrow [R(\lambda p) + T]_{C_2} = [\mu q]_{C_2}$

Or now that we are in the same coordinate system:

$$R(\lambda p) + T = \mu q$$

Two Calibrated Views of the Same 3D Scene



$$R(\lambda p) + T = \mu q$$

Given 2D correspondences (p, q)

Find motion R, T and depths λ, μ .

PnP vs. 2-View Structure from Motion (SfM)

PnP

1. Gn. “world frame” points P_i and corresponding calibrated coordinates $K^{-1}x_i$
2. Set camera frame 3D coordinates (with scale/depth ambiguity) to $\lambda_i K^{-1}x_i$
3. Then solve (for R, T, λ):

$$\lambda_i K^{-1}x_i = R P_i + T \quad \forall i$$

Where R, T denotes transformation between camera and world.

Structure from Motion

1. No “world frame”. Instead, just calibrated image plane coordinates $p_i = K_1^{-1}x_{i1}$ and $q_i = K_2^{-1}x_{i2}$ **of the same 3D point P_i .**
2. So corresponding camera frame 3D coordinates in the two frames are: $\lambda_i p_i$ and $\mu_i q_i$.
3. Now solve for “motion between cameras” R, T and the scales λ_i, μ_i (which permit getting 3D coordinates of P_i in either camera frame)

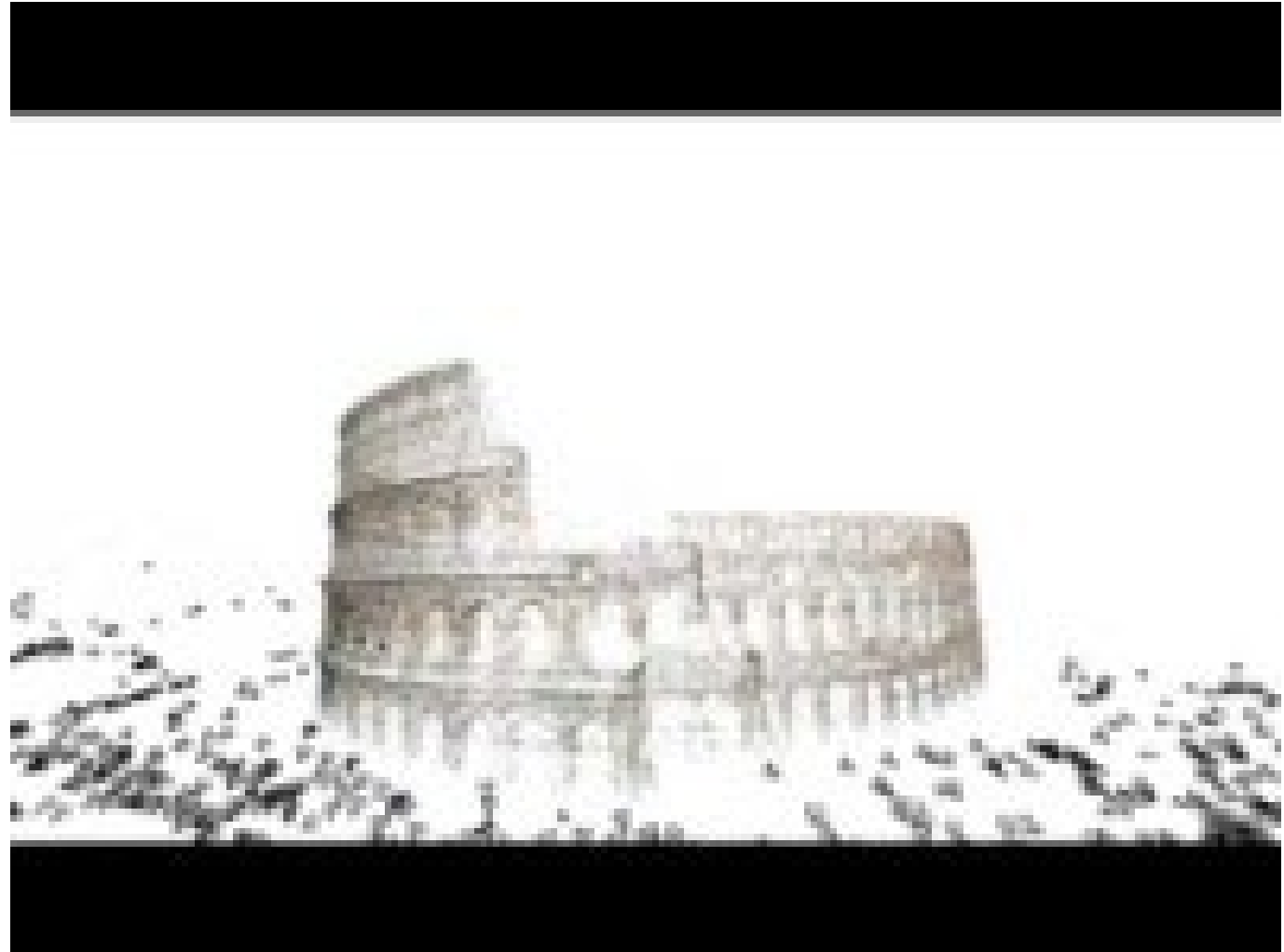
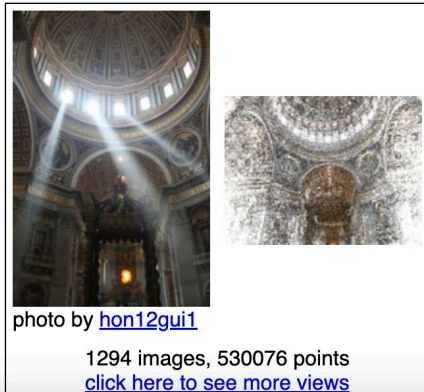
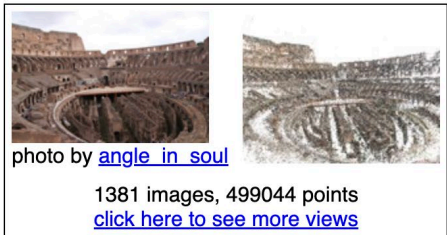
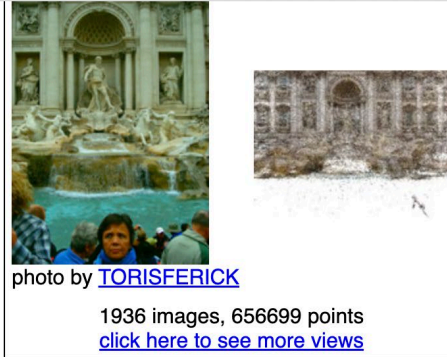
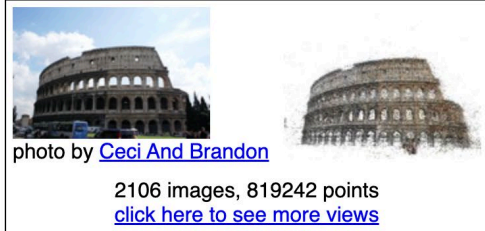
SfM, SLAM, Visual Odometry

- **“SfM” (structure from motion):** graphics and computer vision folks interested mainly in building 3D models of scenes/objects. Images could be unordered, from multiple different cameras taking pictures of the same 3D scene/object. Usually performed offline.
- **“VO” (visual odometry):** mainly interested in localizing and tracking the robot alone, i.e. how far has it traveled in what direction, what speed etc. over short time window. Images usually frames of a video captured from a moving camera in a static / close-to-static world. Often performed online, as images stream in. Usually a single pre-calibrated camera.
- **“SLAM” (simultaneous localization and mapping):** Jointly estimating 3D models of scenes and localizing the robot (camera) w.r.t. that scene. Images similar to VO, also streaming, but emphasis on long-term map consistency. VO++. We'll see an example of a SLAM system later.

Methods largely the same, with differences in emphases depending on which output (depths, transformations, or both) matter most.

SfM Application: Building Rome in a Day

SfM techniques applied to Flickr image collections!

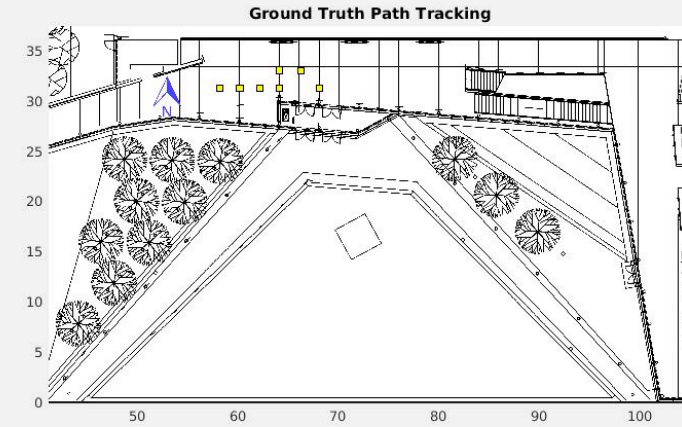
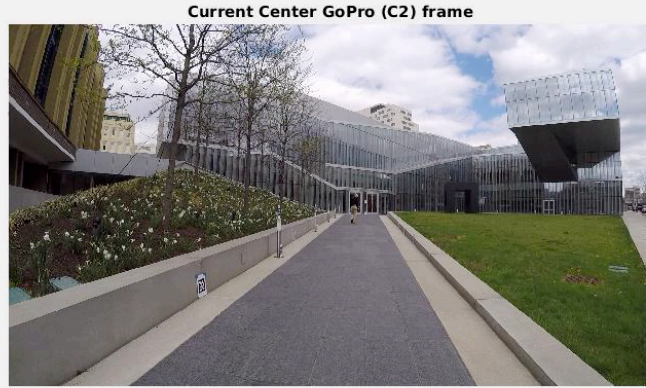


SfM Application: 3D Presidential Portrait



Really MVS
rather than SfM,
since the camera
locations are
fully known.

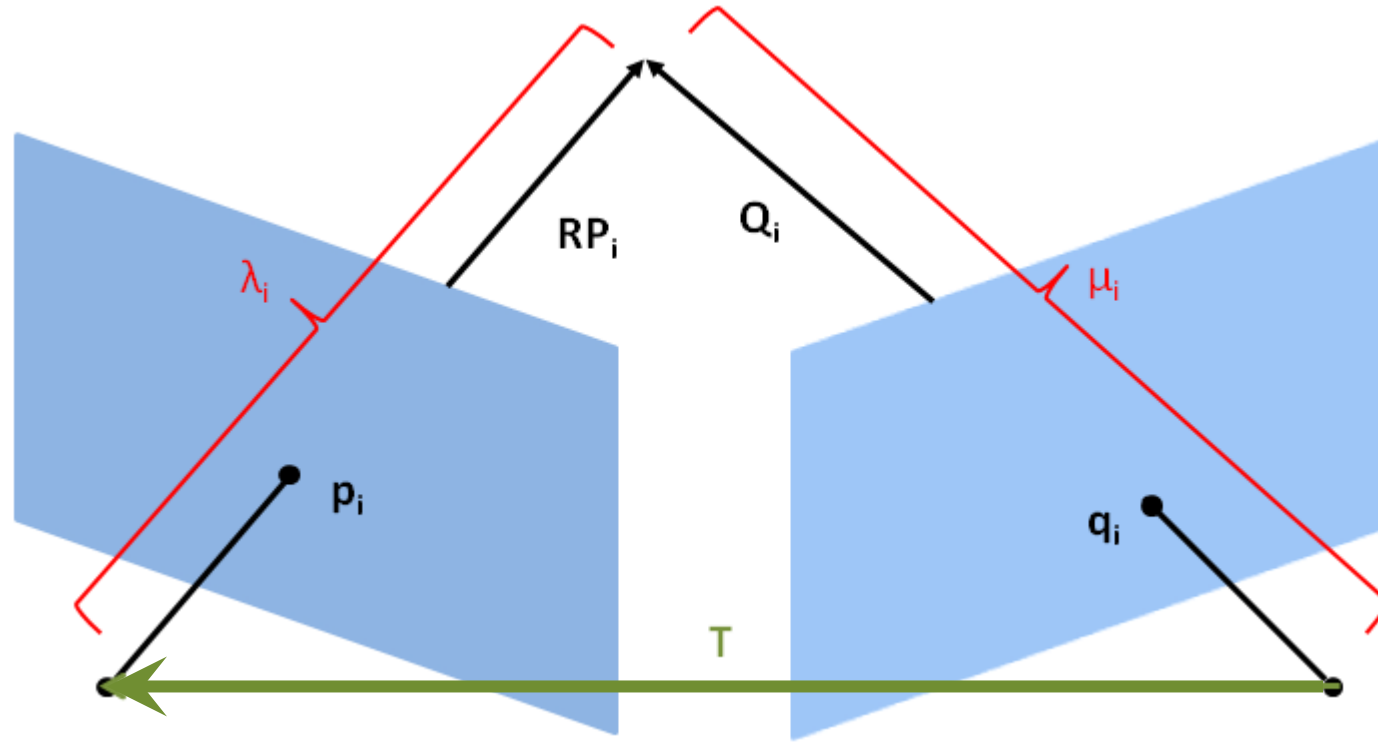
SfM is the main ingredient of Visual Odometry



SLAM for Driving in DARPA Grand Challenge (circa 2007)



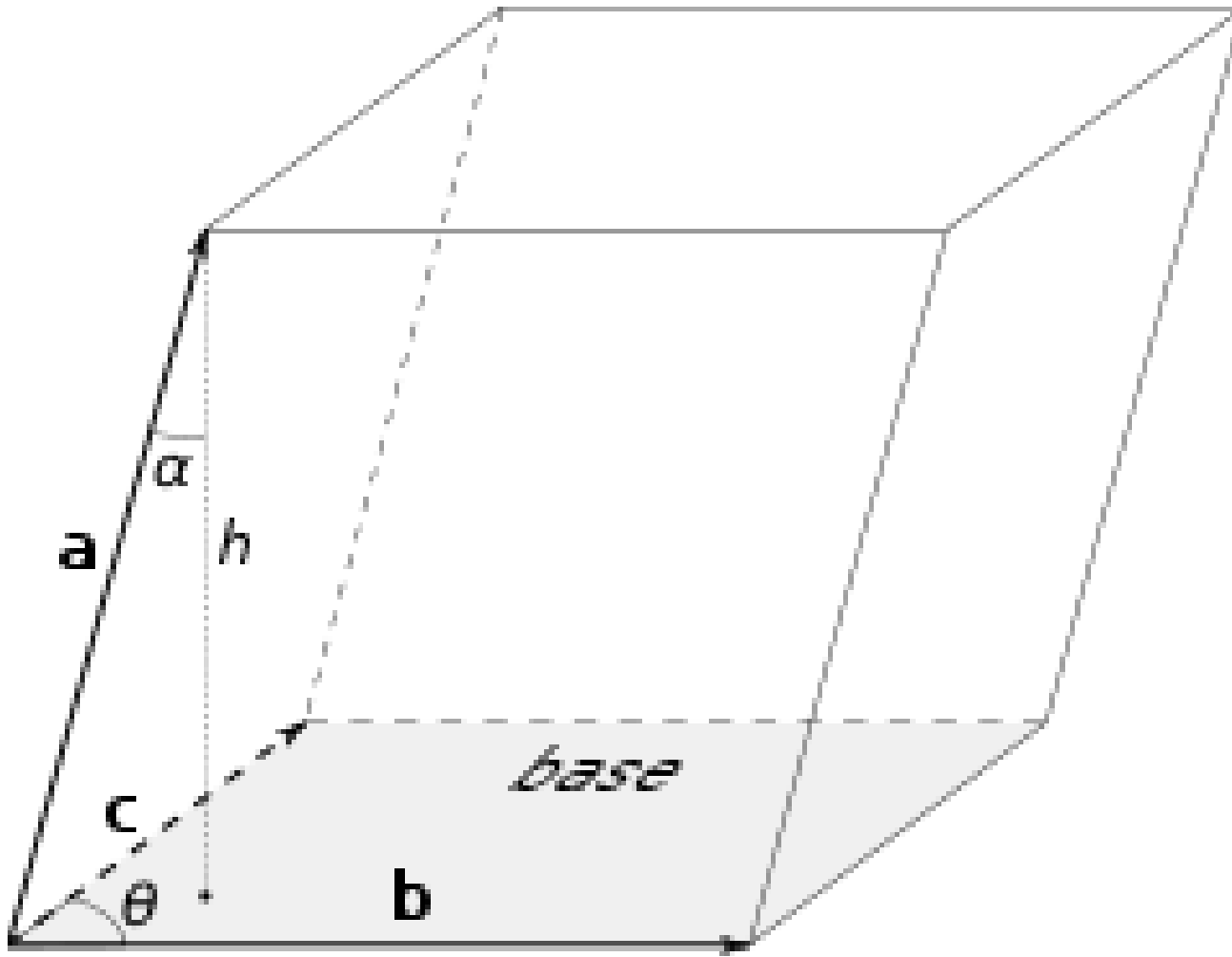
“Epipolar Constraints” Between Two Views of a Scene



We can eliminate the depths from $R(\lambda p) + T = \mu q$ and obtain the epipolar constraint:

$$q_i^T (T \times R p_i) = 0$$

Review: Mixed Product = Volume of Parallelepiped



$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b})$$

In our setting, q, Rp, T are edges of a triangle because

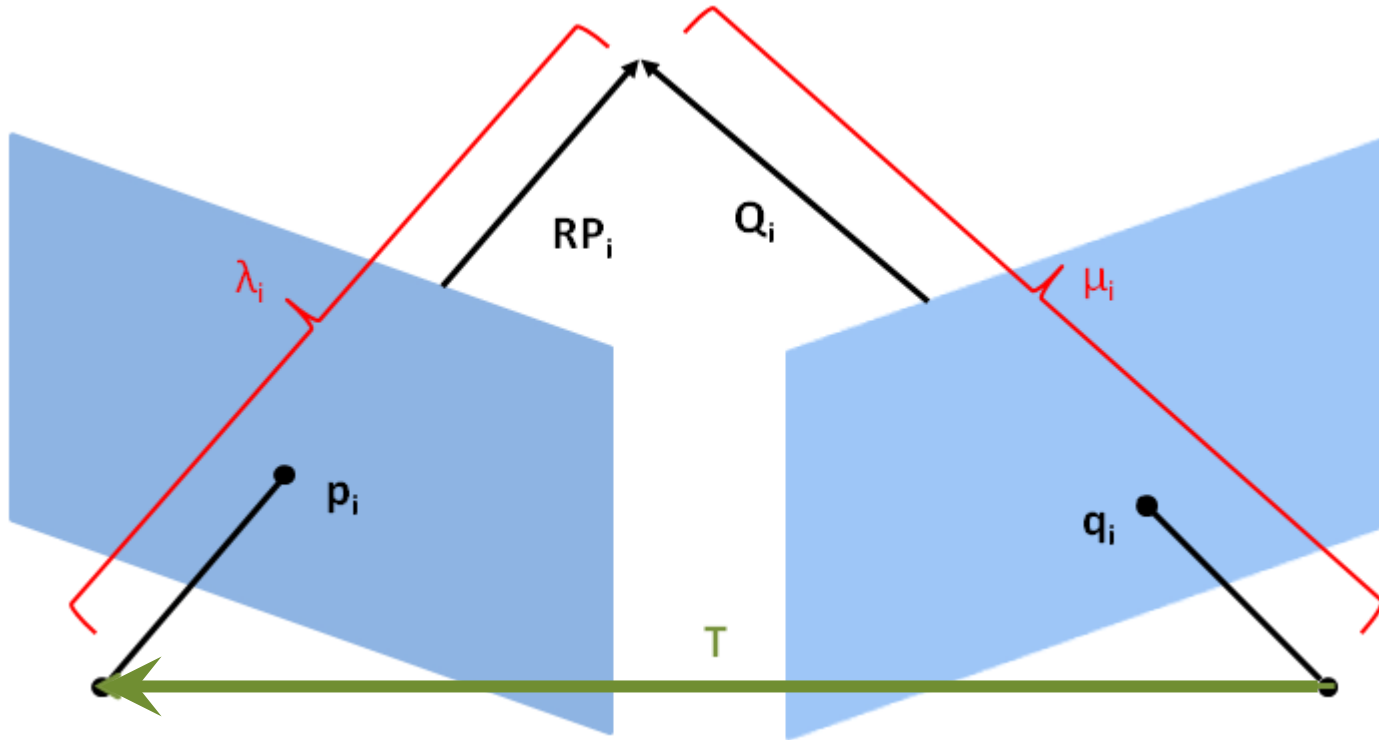
$$\lambda R\mathbf{p} + \mathbf{T} = \mu \mathbf{q}$$

(which is the triangle rule for vectors).

And triangles are planar!

Fun fact: this is also the determinant of a matrix with columns **a**, **b**, **c**.

Geometric Meaning of Epipolar Constraint



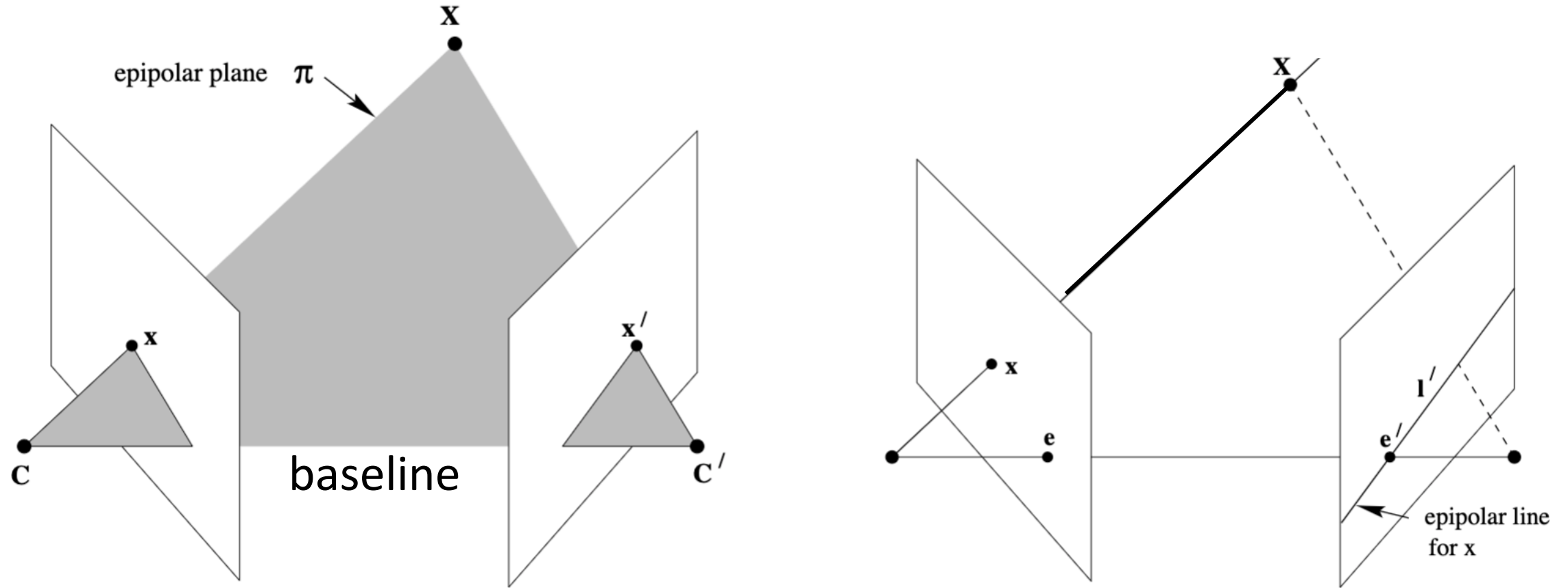
The two rays q and Rp intersect in space if and only if they are coplanar with the translation vector T . Three vectors are coplanar if their mixed product vanishes:

$$q^T(T \times Rp) = 0$$

Epipolar constraint =
“image rays from the world point to the two cameras lies on the same plane as the baseline (translation vector) connecting the camera centers.”

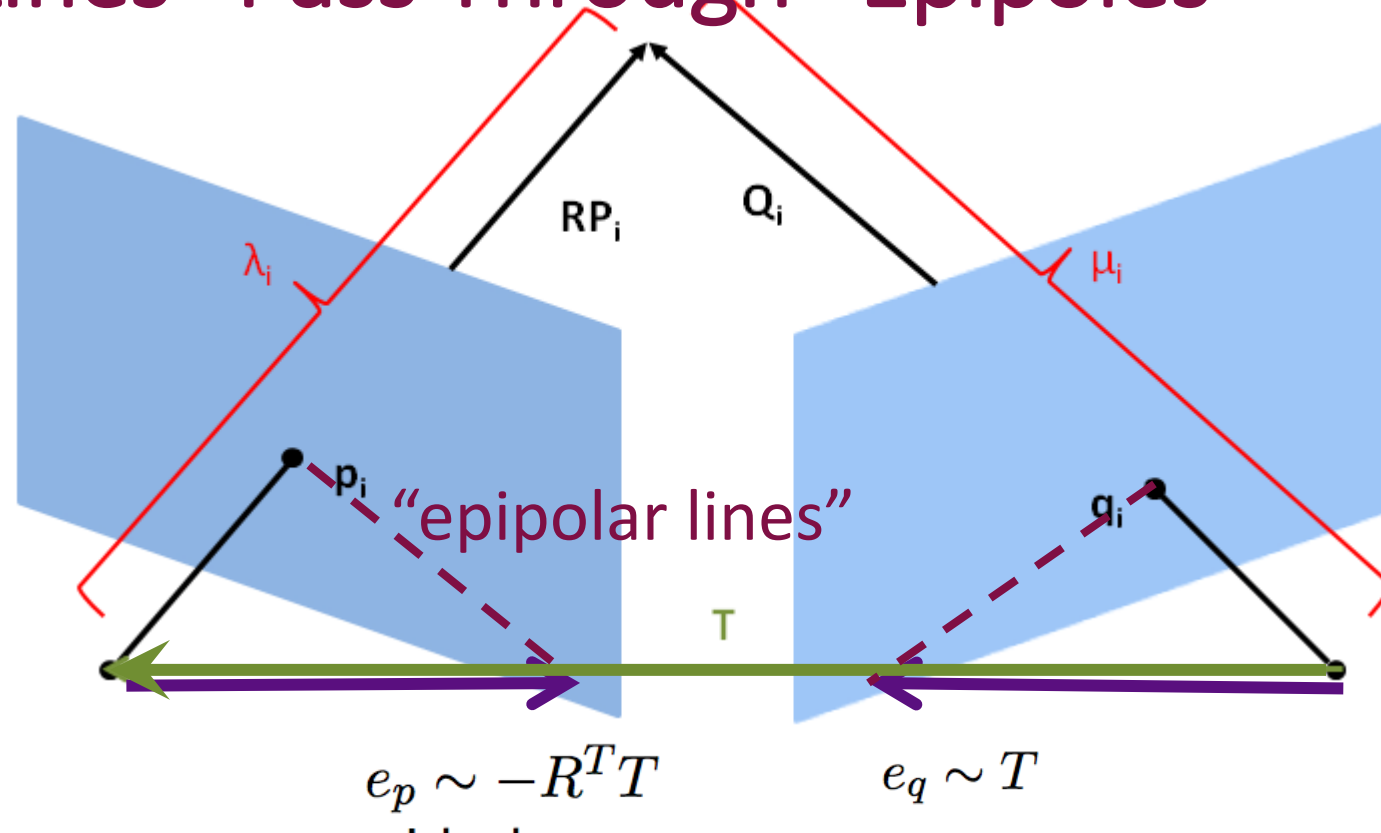
Again, these are just the edges of a triangle with the 2 camera centers and the world point as vertices, so naturally coplanar!

Epipolar “Planes” and “Lines”



Epipolar planes are planes containing the baseline.
Any 3D point induces a corresponding epipolar plane.
Intersection of this plane with an image plane = epipolar line for that 3D point.

“Epipolar Lines” Pass Through “Epipoles”



$e_p \sim -R^T T$ and $e_q \sim T$ are the “epipoles” = images of the other camera center on each plane = intersections of baseline T with the two planes = VP of the translation direction in each plane.

All epipolar lines in each image plane pass through its epipole.

Collecting Epipolar Lines into Epipolar “Pencils”

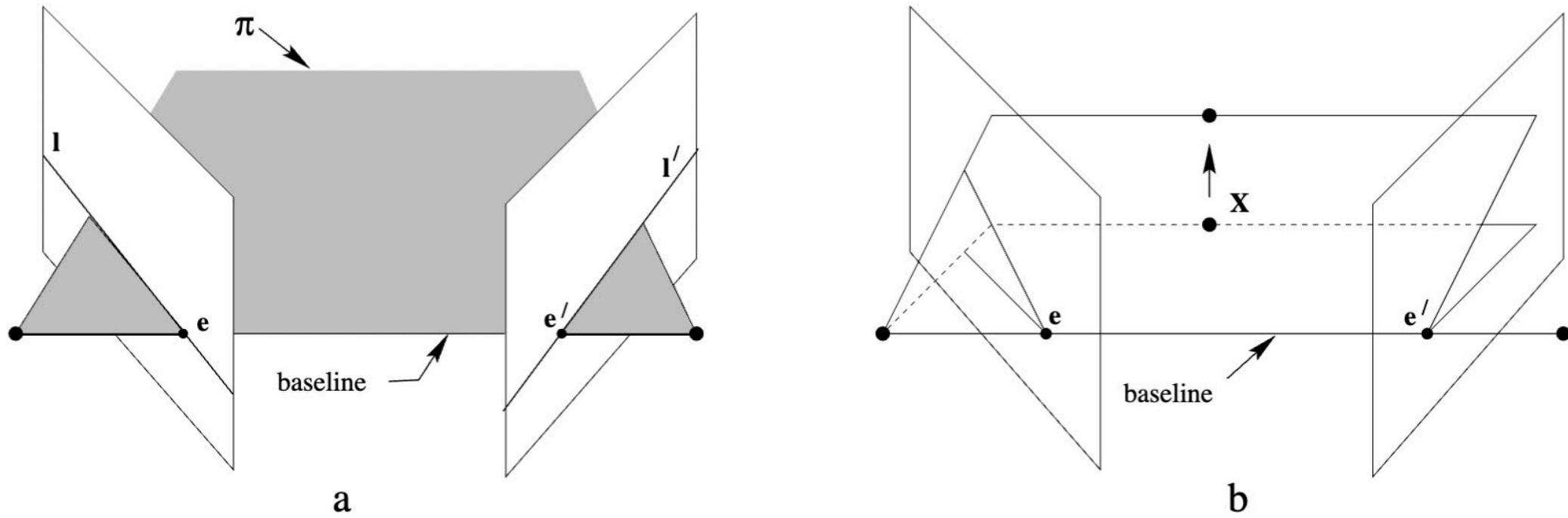
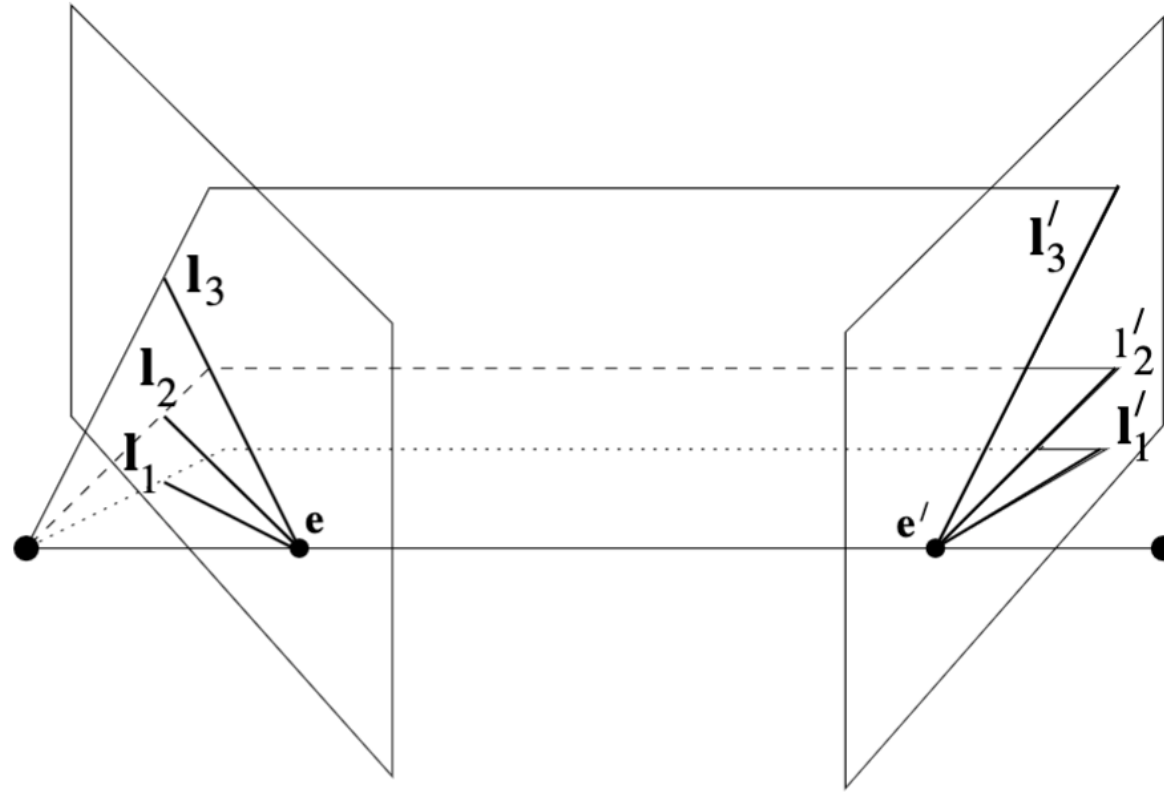


Fig. 9.2. Epipolar geometry. (a) The camera baseline intersects each image plane at the epipoles e and e' . Any plane π containing the baseline is an epipolar plane, and intersects the image planes in corresponding epipolar lines l and l' . (b) As the position of the 3D point X varies, the epipolar planes “rotate” about the baseline. This family of planes is known as an epipolar pencil. All epipolar lines intersect at the epipole.

Epipolar “Pencils”



The epipolar lines in each image plane form a “pencil” whose tip is the epipole i.e. image of the other camera center / intersection of baseline with image plane. (Because all epipolar planes contain the baseline, after all)

Epipolar Pencils of Frontoparallel Cameras

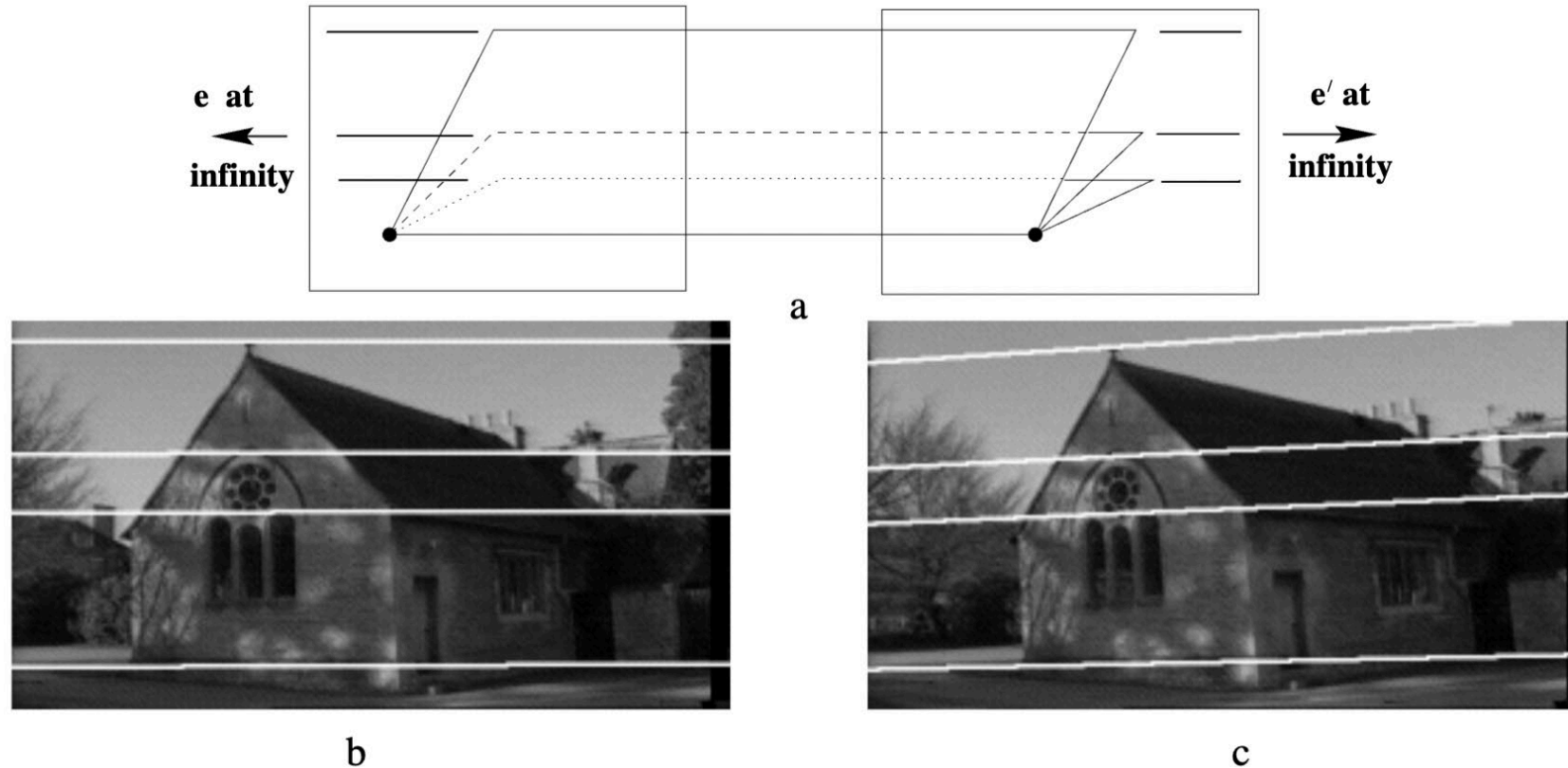
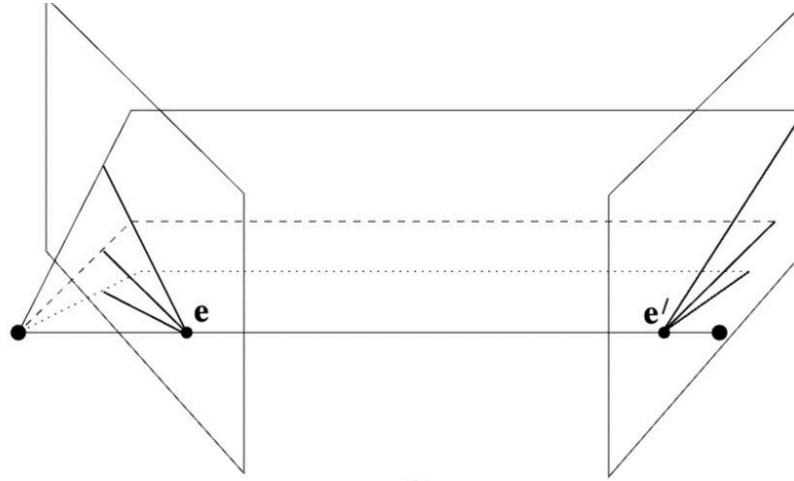


Fig. 9.4. **Motion parallel to the image plane.** In the case of a special motion where the translation is parallel to the image plane, and the rotation axis is perpendicular to the image plane, the intersection of the baseline with the image plane is at infinity. Consequently the epipoles are at infinity, and epipolar lines are parallel. (a) Epipolar geometry for motion parallel to the image plane. (b) and (c) a pair of images for which the motion between views is (approximately) a translation parallel to the x -axis, with no rotation. Four corresponding epipolar lines are superimposed in white. Note that corresponding points lie on corresponding epipolar lines.

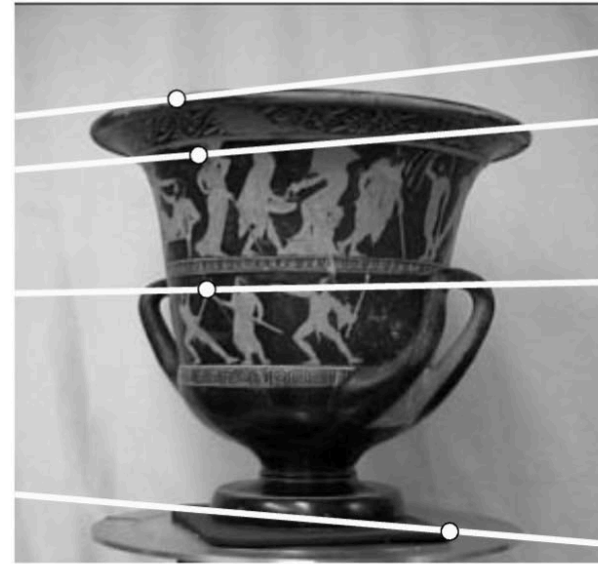
Epipolar Pencils visually identify camera orientations



a



b



c

Epipolar Pencils visually identify camera orientations

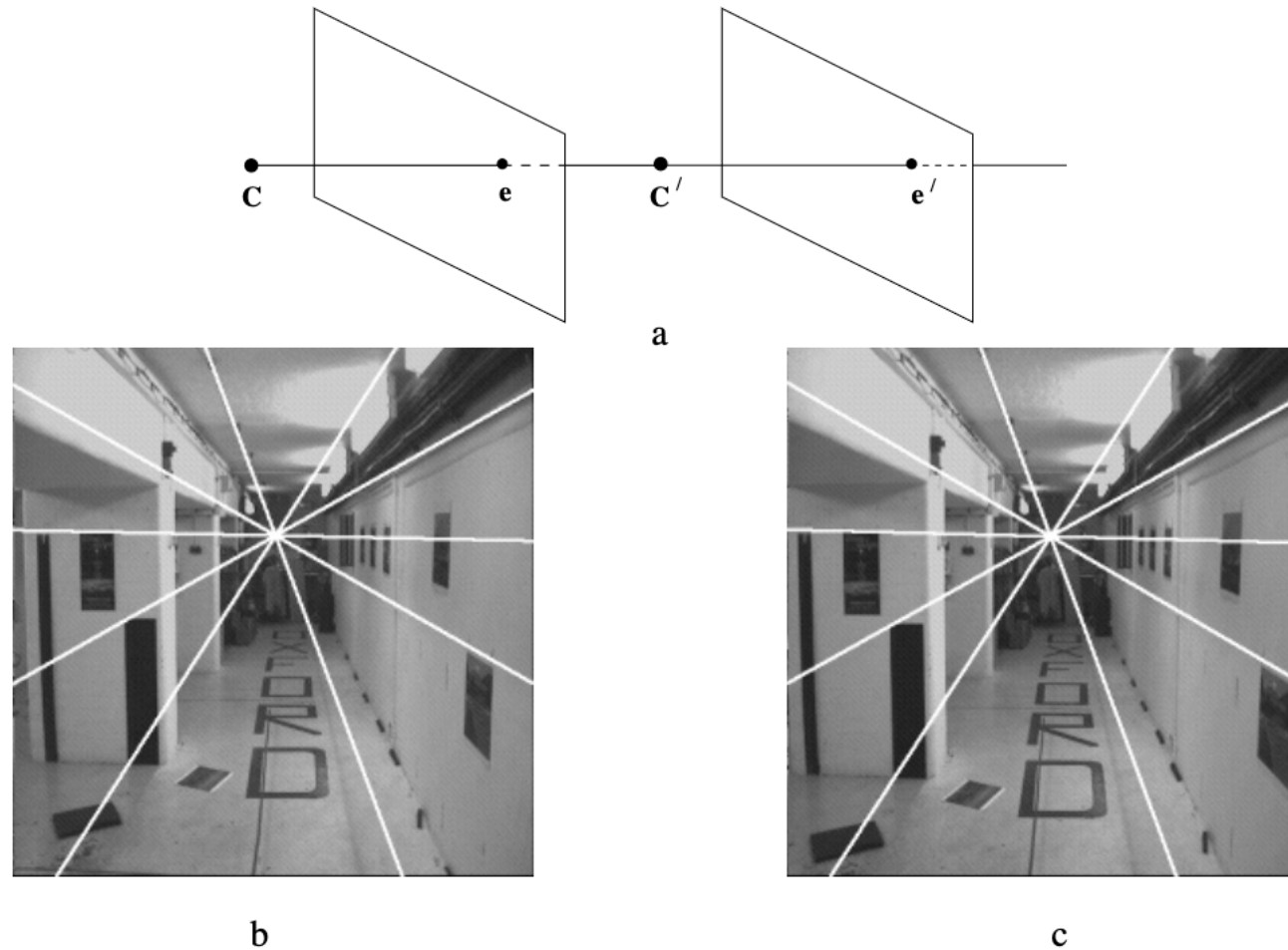


Fig. 9.8. **Pure translational motion.** (a) under the motion the epipole is a fixed point, i.e. has the same coordinates in both images, and points appear to move along lines radiating from the epipole. The epipole in this case is termed the Focus of Expansion (FOE). (b) and (c) the same epipolar lines are overlaid in both cases. Note the motion of the posters on the wall which slide along the epipolar line.

Summary so far (Essential matrix and Epipolar Stuff)

- Essential matrix contains information about R, T between two views, provides “epipolar constraint” equations that might help identify E and eventually solve the SfM problem.
- Geometric intuitions connected to epipoles, epipolar planes, lines, pencils. Just visualizing epipolar pencils already gives us qualitative information about the relative orientations of the two camera views, hinting at the utility of these concepts for SfM.
- Epipolar lines directly constrain point correspondences, will be useful for later on in the course.

